

Turn Electricity Into Music!



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How Does Music Come Out of a Speaker?

- Build Your Own Paper Speaker!

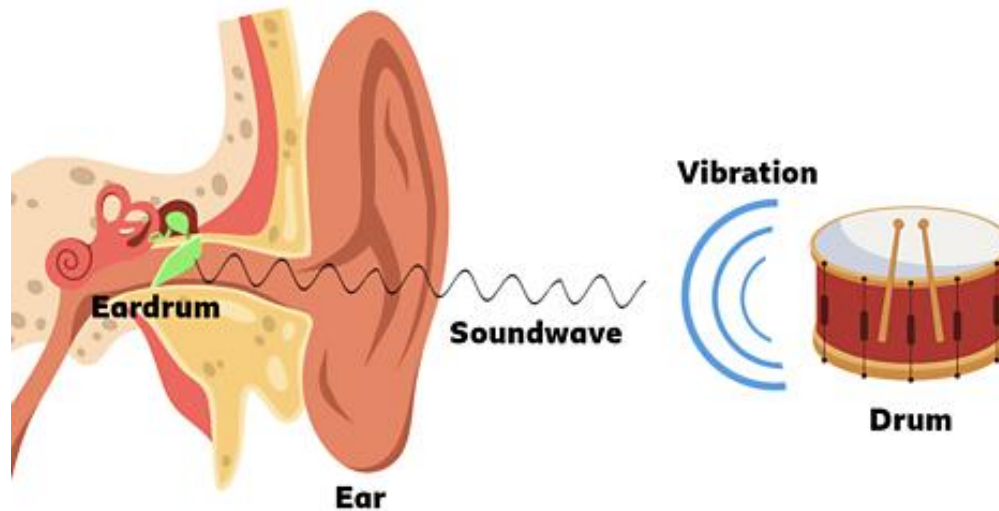


Is There a Tiny Singer Inside Your Phone? 🤔



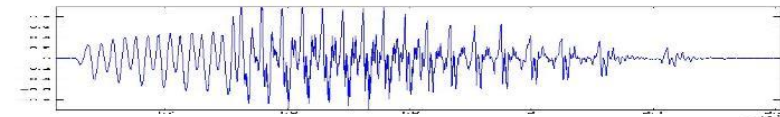
How does electricity become sound?

What Is Sound? 🎵



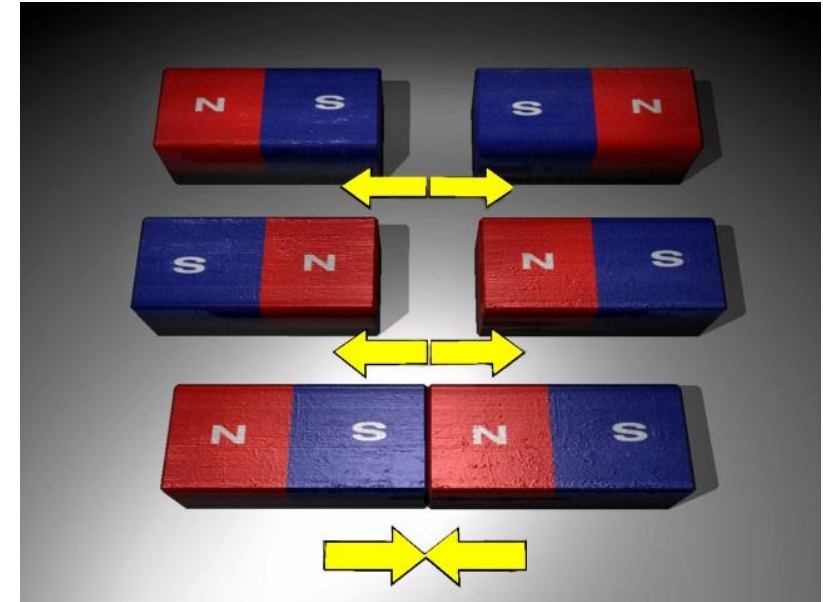
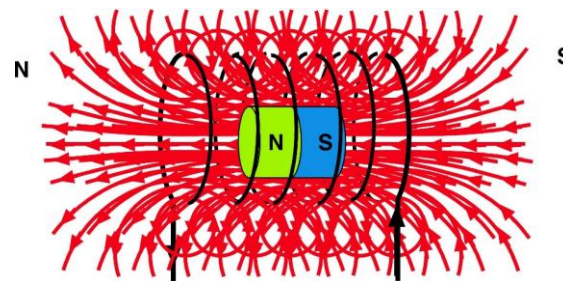
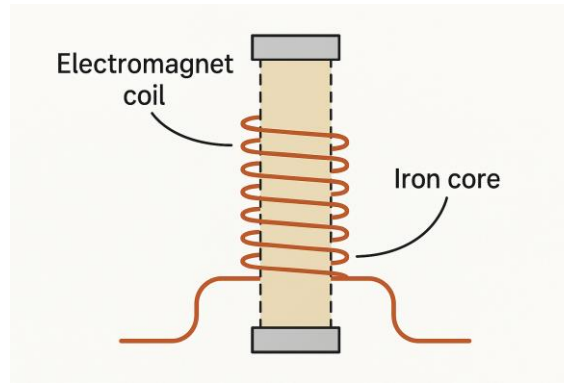
- **Something vibrates**
- **Air moves back and forth**
- **Your ears hear the vibration**

- We must convince the paper to vibrate in synch with our music
- Our music is generated by our (stereo, mp3player, etc.) as an electrical signal
- High frequency electrical signal=high frequency sound
 - Low = Low
 - Music is a combination of lots of frequencies



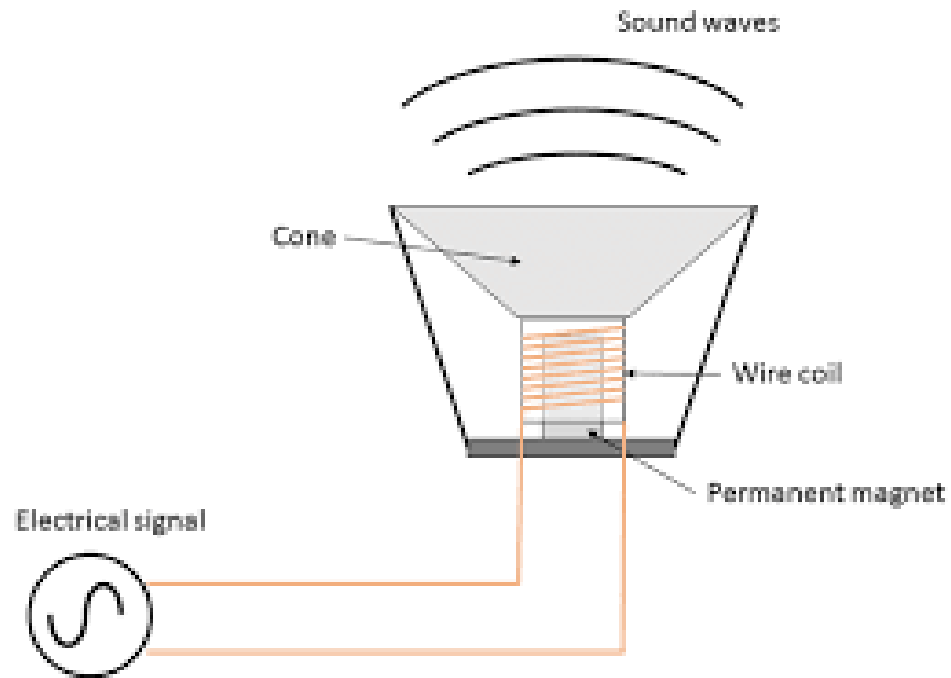
The Secret Power: Magnets

- Electricity flows in the coil
- The coil becomes a magnet
- It pushes and pulls the speaker magnet



The wire coil becomes a magnet with north and south sides. When the music signal changes, those sides switch. That makes the coil get pushed and pulled by the permanent magnet.

How a Speaker Works 🗣️



- 1. Electricity enters the coil**
- 2. Magnet makes the cone move**
- 3. Cone pushes air = sound**

Let's Build Our Own Speaker! 🛠️ 🎵

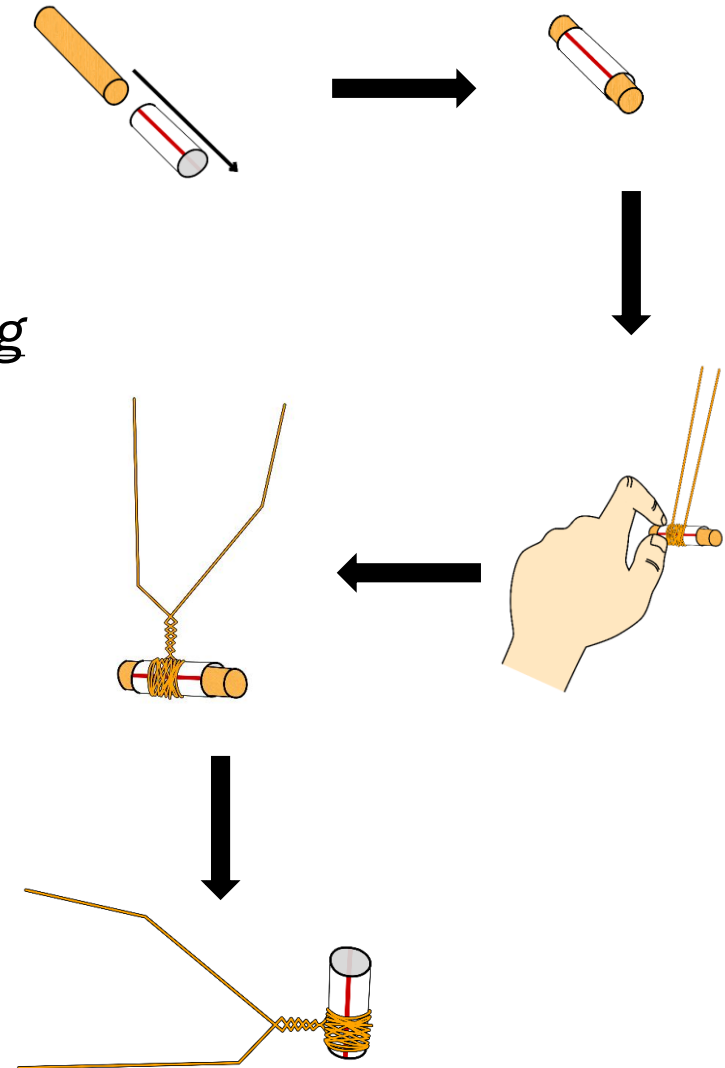
- Paper cone
- Copper wire coil
- Magnet
- Phone or music source



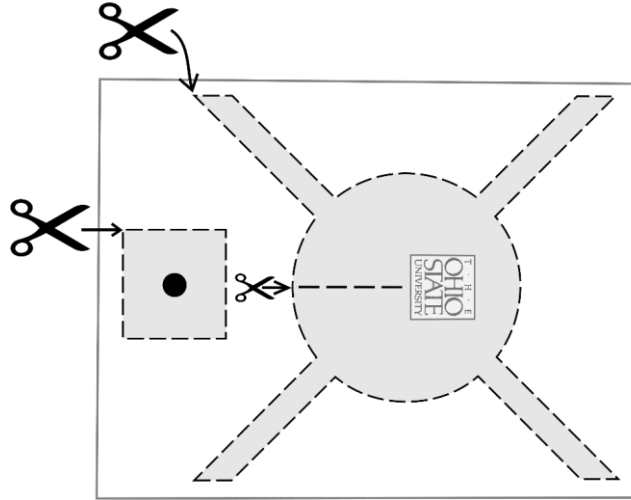
Step 1: Make the Wire Coil

- Slide straw over dowel
- Wrap wire around straw
 - Leave about 6" free before you start winding
 - Leave 6" at the end
- Overlap wire as much as possible
 - Make sure all wire is on the straw
- Twist wire so it doesn't unwind
- When finished, slide coil to one end of straw
- Remove dowel

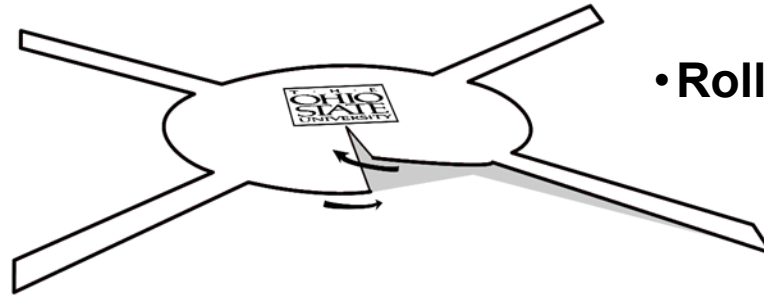
Burn 1 inch of the end of each of the two wires coming off the coil.



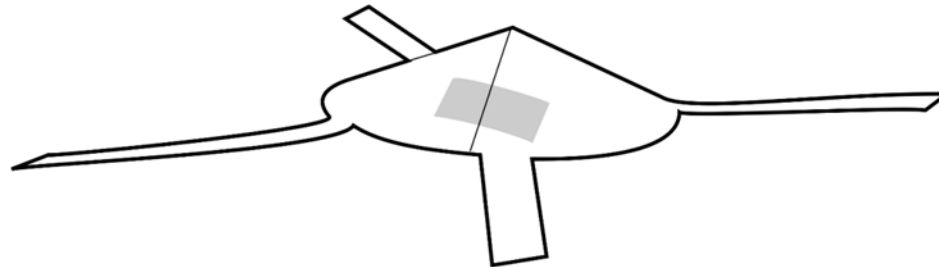
Step 2: Make the Paper Cone



• Cut the paper circle



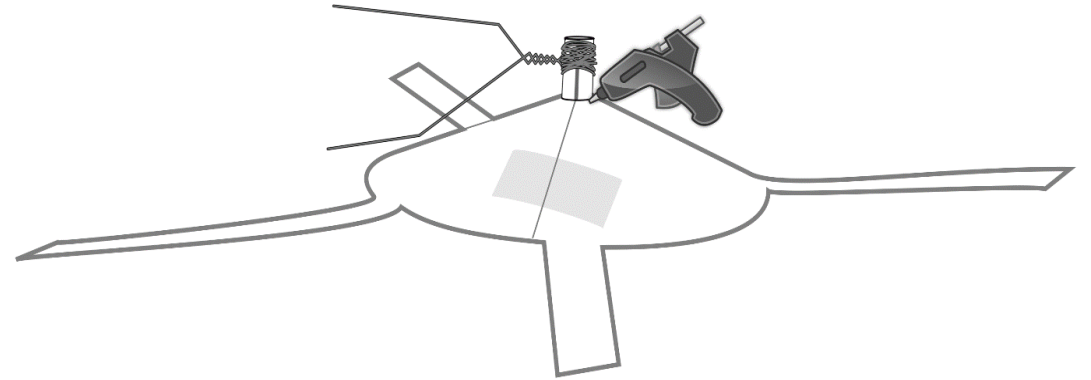
• Roll it into a cone



• Tape the edge

Step 3: Attach the Coil to the Cone 🔗 🔊

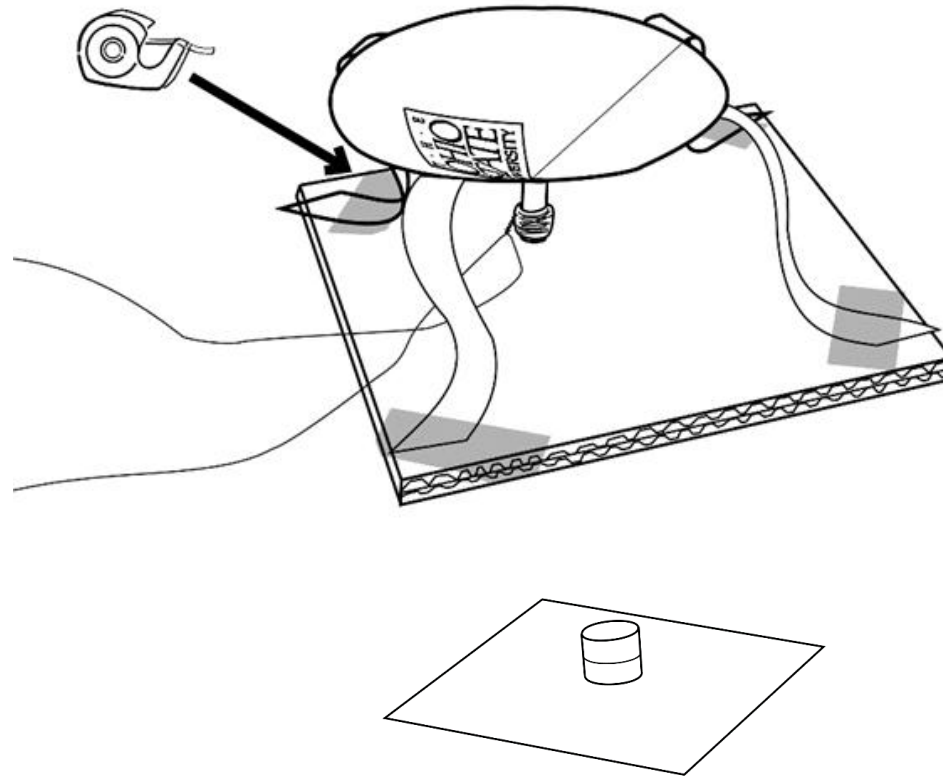
- Glue long end of straw to bottom of cone
 - Coil should be at end away from cone
- Don't get glue on the end of the straw away from the cone – need room for magnet to move!



- **Glue the coil to the cone**
- **Keep the coil centered**
- **Let it dry**

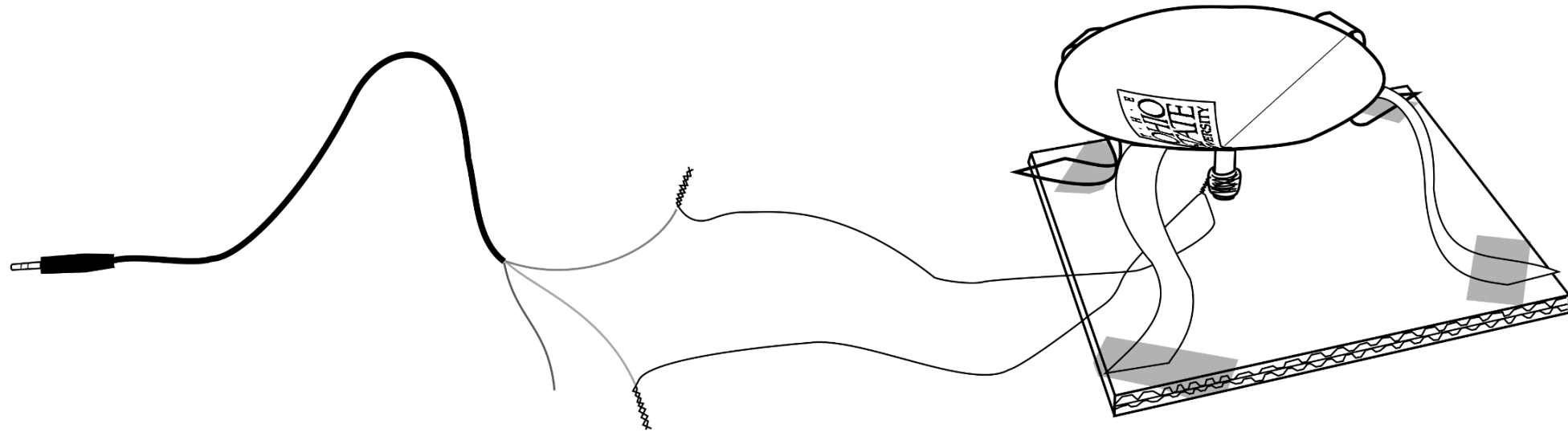
Step 4: Add the Magnet

- Place the magnet inside the coil
- Make sure it can move freely
- Do not glue the coil to the magnet



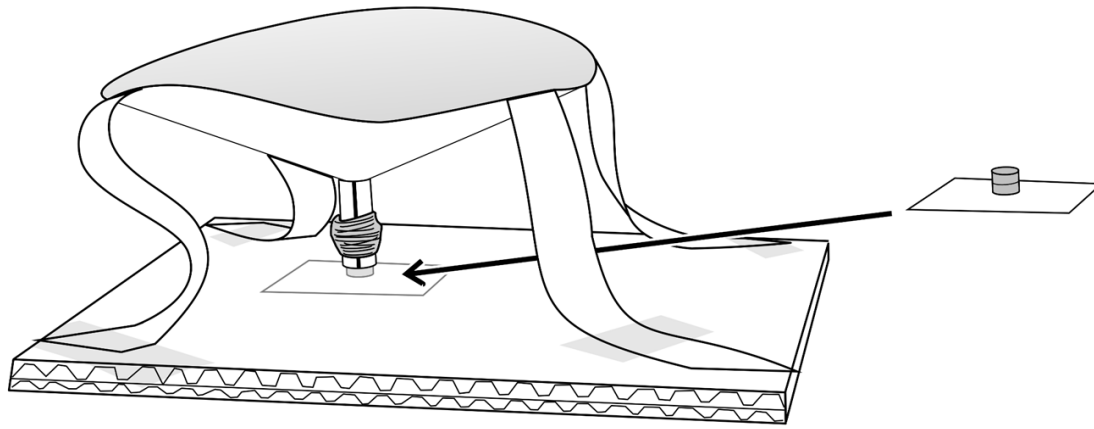
Coil should hover above or gently rest on cardboard after all four legs have been taped.

Step 5: Connect to audio cable 🎵

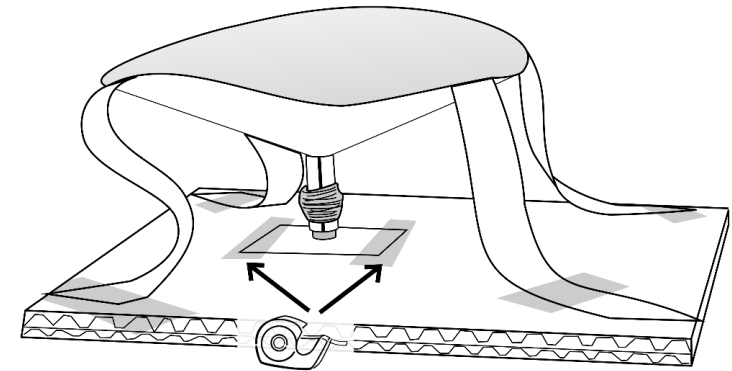


- Twist yellow wire and one lead from your speaker together.
- Twist red or white wire with other speaker lead.
- Tape the black audio cable to the edge of your cardboard.

Note: Position Magnets Inside Coil

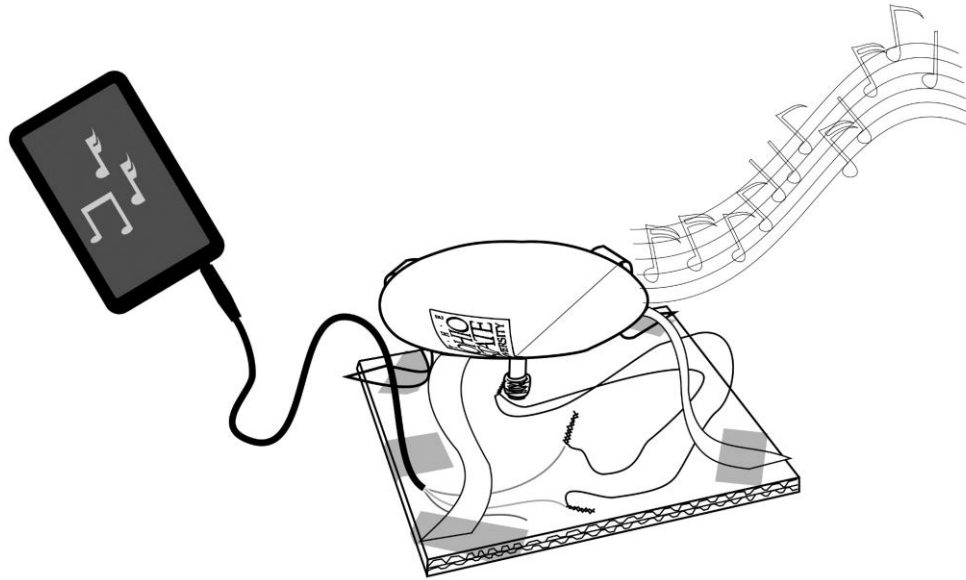


Do not tape paper down! (Yet)



Once it works, tape the square

Step 5: Connect Your Phone 🎵 📱



- **Connect the wire to your phone**
- **Play music**
- **Listen for the sound**

You have applied electromagnetism to create sound from an electrical signal.

Troubleshooting

- Volume set to max?
- Speaker is plugged into phone all the way?
- Magnets are inside the coil?
- Coil can easily slide up and down on the magnets?
- Coil is at the bottom of the straw? (Closest to the cardboard)
- 1 inch of each speaker lead was burned?
- The two twisted connections are not touching each other?
- If your audio cable doesn't have a red, white and yellow wire, you can use any two wires
- Try re-twisting the audio cable to the speaker leads!

Where Do We Use This in Real Life?



- Phones
- Headphones
- Cars
- Smart speakers

